

Functional Requirements Document (FRD)

Customer Lifetime Value & Churn Segmentation Analysis

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Related Document	Business Requirements Document (BRD)
Project	Customer Lifetime Value & Churn Segmentation Analysis

1. Purpose

This document translates the business requirements defined in the BRD into specific functional requirements for the data pipeline, analysis, and dashboard.

2. Data Requirements

- System shall source order, customer, payment, and product data from the transaction dataset.
- System shall exclude orders that are not in "delivered" status from all value and recency calculations.
- System shall use the customer's unique identifier (not the per-order identifier) to correctly group repeat purchases to the same person.

3. Functional Requirements

ID	Requirement	Description
FR-1	RFM Scoring	System shall calculate Recency, Frequency, and Monetary scores per customer on a 1–5 scale.
FR-2	Customer Segmentation	System shall assign each customer to one of eight defined segments based on their R, F, and M scores.
FR-3	Churn Definition	System shall flag a customer as churned if no purchase has occurred within 90 days of their last order.
FR-4	Churn Prediction	System shall estimate a churn probability per customer using a classification model, excluding recency from model features to avoid label leakage.
FR-5	Segment Revenue & Distribution View	Dashboard shall show total money spent by segment and percentage of customers by segment.
FR-6	Average Revenue per Customer View	Dashboard shall show average money spent per customer, by segment.
FR-7	Recency Distribution View	Dashboard shall display the distribution of customer recency (days since last order) in bins, with the 90-day churn threshold visually marked.
FR-8	At-Risk Action List	Dashboard shall display a table of At-Risk (High Value) customers with money spent and churn probability, sorted descending by monetary value.
FR-9	Active Customers Trend View	Dashboard shall show the trend of active customer count by year and month.
FR-10	Cohort Retention View	Dashboard shall display customer retention by acquisition cohort over time.
FR-11	Filtering	Dashboard shall allow filtering all visuals by year/month and by churned status.

4. Non-Functional Requirements

- Dashboard visuals should load and respond to filter selections without significant delay.
- Underlying SQL queries should be indexed appropriately to support acceptable performance on the full dataset.
- Segment logic should be transparent and explainable to a non-technical business stakeholder.

5. Assumptions

- The 90-day churn threshold is configurable and may be revisited based on business input.
- The dashboard is intended for internal business use, not customer-facing distribution.

6. Acceptance Criteria

- Every customer in the dataset is assigned to exactly one segment.
- The at-risk customer list can be filtered to show only high-value, high-churn-probability customers.
- Dashboard filters (year/month, churned status) update all relevant visuals correctly when applied.
- Recency distribution view visually marks the 90-day churn threshold.